

Python assignment - Answers

Python assignment

1. Load the Customer data sheet in Python. Set churned as y. Remove churned and customer_id from X.

Answer: Read the Customer data sheet, assign churned to y, and exclude both churned and customer_id from X. The remaining ten columns are predictors.

2. Make a stratified 80/20 training and test split. Check how many customers canceled in each split.

Answer: Use `train_test_split(test_size=0.20, stratify=y, random_state=42)`. The source workbook has 500 records: 360 stayed and 140 churned. Expected split counts are train: 288 stayed/112 churned; test: 72 stayed/28 churned.

3. Train these five models separately: KNN, Gaussian Naive Bayes, Decision Tree, Random Forest, and Logistic Regression.

Answer: Fit `KNeighborsClassifier`, `GaussianNB`, `DecisionTreeClassifier`, `RandomForestClassifier`, and `LogisticRegression` as separate estimators.

4. Use a StandardScaler pipeline for KNN and Logistic Regression. Fit the scaler using training data only.

Answer: Put `StandardScaler` inside a Pipeline before KNN and Logistic Regression. Pipeline fitting learns the scaler from `X_train` only and applies those training statistics to test rows.

5. For each model, report test accuracy, precision, recall, F1 -score, and a confusion matrix.

Answer: For each model, compute accuracy, precision, recall, and F1 with positive label 1, plus `confusion_matrix(y_test, predictions, labels=[0,1])`. The accompanying notebook prints the exact scores and matrices.

6. Compare the models: Which catches the most customers who cancel? Which produces the fewest false alarms? Would you choose the same model for both goals?

Answer: The model with highest churn-class recall catches the most customers who cancel (fewest false negatives). The model with the fewest false positives creates the fewest false alarms. These can be different models; compare the notebook's measured table rather than assuming one winner.

7. Inspect what each model learned. For example, compare K values, display a small decision tree, examine forest feature importance, and inspect logistic regression coefficients.

Answer: Tune K by cross-validation on training data, print a shallow tree, rank Random Forest `feature_importances_`, and inspect Logistic Regression coefficients after scaling.

8. Write a short recommendation explaining which model you would use to identify customers for a retention campaign.

Answer: For retention outreach, prioritize a validated model with high churn recall while maintaining precision that the campaign can support. Select the operating threshold using the relative costs of missed churners and unnecessary contacts; do not choose by accuracy alone. This workbook is synthetic classroom data and should not drive real customer actions.